

# Practical · Large and small units of length and area

Cambridge insert: the conversions that decide whether an answer is right or a hundred times wrong.

LESSON 59

1 × 40 MIN

GEOMETRY 8, TEMA 60

STAGE 9 · 7.3

LESSON CLOCK

5'

12'

14'

5'

4'

|                                 |        |
|---------------------------------|--------|
| Guess the area of the classroom | 5 min  |
| Length conversions              | 12 min |
| Area conversions                | 14 min |
| Scale and estimation            | 5 min  |
| Homework                        | 4 min  |

LEARNING OBJECTIVES

- Convert between units of length and of area.
- Explain why area conversions use the square of the length factor.
- Choose a sensible unit for a given measurement.
- Estimate and check answers for plausibility.

TEXTBOOK

|           |                            |
|-----------|----------------------------|
| National  | Tema 60, pp. 144–146       |
| Cambridge | Learner’s Book pp. 156–162 |
| Workbook  | Workbook 7.3               |

01

## Explanation

### Why area factors are squared

A square of side  $1\text{ m}$  is also a square of side  $100\text{ cm}$ . Its area is  $1\text{ m}^2$  — and also  $100 \cdot 100 = 10\,000\text{ cm}^2$ . The length factor is  $100$ ; the area factor is  $100^2 = 10\,000$ .

THE RULE

If lengths are multiplied by  $k$ , areas are multiplied by  $k^2$  (and volumes by  $k^3$ ).

| LENGTH                                           | FACTOR   | AREA                                     | FACTOR    |
|--------------------------------------------------|----------|------------------------------------------|-----------|
| $1\text{ cm} = 10\text{ mm}$                     | $10$     | $1\text{ cm}^2 = 100\text{ mm}^2$        | $100$     |
| $1\text{ m} = 100\text{ cm}$                     | $100$    | $1\text{ m}^2 = 10\,000\text{ cm}^2$     | $10\,000$ |
| $1\text{ km} = 1\,000\text{ m}$                  | $1\,000$ | $1\text{ km}^2 = 1\,000\,000\text{ m}^2$ | $10^6$    |
| $1\text{ ha} = 100\text{ m} \times 100\text{ m}$ | —        | $1\text{ ha} = 10\,000\text{ m}^2$       | —         |

THE COMMONEST ERROR IN THE WHOLE YEAR

Writing  $1\text{ m}^2 = 100\text{ cm}^2$ . It is  $10\ 000$ . When in doubt, draw the square and count.

Scale and estimation

A map at scale  $1 : 50\ 000$  shrinks every length by  $50\ 000$ , so it shrinks every area by  $50\ 000^2 = 2.5 \times 10^9$ . One square centimetre on that map represents  $0.25\text{ km}^2$ .

Before calculating anything, decide what a sensible answer looks like:

| THING             | ROUGHLY                   |
|-------------------|---------------------------|
| a page of a book  | $\approx 600\text{ cm}^2$ |
| a classroom floor | $\approx 50\text{ m}^2$   |
| a football pitch  | $\approx 0.7\text{ ha}$   |
| a small district  | $\approx 100\text{ km}^2$ |

If a calculation gives a classroom of  $5000\text{ m}^2$ , the arithmetic is not wrong by a little — a unit was converted the wrong way.

02

Worked examples

EXAMPLE 1 Convert  $4.5\text{ m}^2$  to  $\text{cm}^2$ , and  $250\ 000\text{ cm}^2$  to  $\text{m}^2$ .

1

$1\text{ m}^2 = 10\ 000\text{ cm}^2$

2

$4.5 \cdot 10\ 000 = 45\ 000\text{ cm}^2$

3

$250\ 000 \div 10\ 000$   
Divide going the other way.

4

$= 25\text{ m}^2$

ANSWER

$45\ 000\text{ cm}^2$  and  $25\text{ m}^2$

**EXAMPLE 2** A field is  $0.8\text{ km}$  by  $0.5\text{ km}$ . Find its area in  $\text{km}^2$ ,  $\text{m}^2$  and hectares.

①  $0.8 \cdot 0.5 = 0.4\text{ km}^2$

②  $0.4 \cdot 1\,000\,000 = 400\,000\text{ m}^2$

③  $400\,000 \div 10\,000 = 40\text{ ha}$

ANSWER

$0.4\text{ km}^2 = 400\,000\text{ m}^2 = 40\text{ ha}$

**EXAMPLE 3** On a map of scale  $1 : 25\,000$ , a lake covers  $8\text{ cm}^2$ . Find its real area in hectares.

① length factor  $25\,000$

② area factor  $25\,000^2 = 625\,000\,000$

③  $8\text{ cm}^2 \cdot 625\,000\,000 = 5 \times 10^9\text{ cm}^2$

④  $= 500\,000\text{ m}^2 = 50\text{ ha}$

$10\,000\text{ cm}^2 = 1\text{ m}^2.$

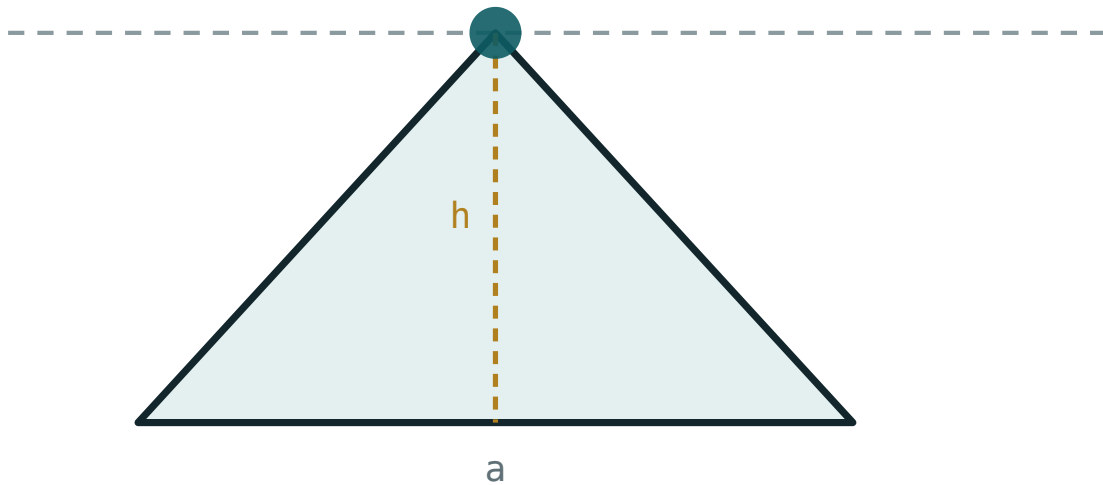
ANSWER

$50\text{ ha}$

### 03 Interactive model

Change the dimensions and watch the area — then convert the readout to another unit by hand.

## Dimensions and area

base a **220**height h **120**area  $\frac{1}{2}ah$  **13200**perimeter **545.6**

### 04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

| ENGLISH            | O'ZBEKCHA                  | РУССКИЙ              |
|--------------------|----------------------------|----------------------|
| Unit of length     | Uzunlik birligi            | Единица длины        |
| Unit of area       | Yuza birligi               | Единица площади      |
| Conversion factor  | Almashtirish koeffitsienti | Коэффициент перевода |
| Millimetre         | Millimetr                  | Миллиметр            |
| Hectare            | Gektar                     | Гектар               |
| Square kilometre   | Kvadrat kilometr           | Квадратный километр  |
| Scale              | Masshtab                   | Масштаб              |
| Estimate           | Taxminlash                 | Оценка               |
| Order of magnitude | Kattalik tartibi           | Порядок величины     |

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

**05 Quick check****Check yourself**

1 m<sup>2</sup> equals:

100 cm<sup>2</sup>

1000 cm<sup>2</sup>

10 000 cm<sup>2</sup>

100 000 cm<sup>2</sup>

1 ha equals:

100 m<sup>2</sup>

1000 m<sup>2</sup>

10 000 m<sup>2</sup>

1 km<sup>2</sup>

3 cm<sup>2</sup> equals:

30 mm<sup>2</sup>

300 mm<sup>2</sup>

3000 mm<sup>2</sup>

0.3 mm<sup>2</sup>

If all lengths are tripled, areas are multiplied by:

2.5 km<sup>2</sup> in hectares is:

## 06 Practice · 21 problems

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Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Convert  $2 \text{ m}^2$  to  $\text{cm}^2$ .

ANSWER  $20\,000 \text{ cm}^2$

2. Convert  $50\,000 \text{ cm}^2$  to  $\text{m}^2$ .

ANSWER  $5 \text{ m}^2$

3. Convert  $4 \text{ cm}^2$  to  $\text{mm}^2$ .

ANSWER  $400 \text{ mm}^2$

4. Convert  $3 \text{ ha}$  to  $\text{m}^2$ .

ANSWER  $30\,000 \text{ m}^2$

5. Convert  $2 \text{ km}^2$  to  $\text{m}^2$ .

ANSWER  $2\,000\,000 \text{ m}^2$

6. Convert  $6000 \text{ mm}^2$  to  $\text{cm}^2$ .

ANSWER  $60 \text{ cm}^2$

7. Convert  $70\,000 \text{ m}^2$  to hectares.

ANSWER  $7 \text{ ha}$

**Medium · the standard question**

7 PROBLEMS



1. Convert  $0.75 \text{ m}^2$  to  $\text{cm}^2$ .

ANSWER  $7500 \text{ cm}^2$

2. Convert  $1.2 \text{ km}^2$  to hectares.

ANSWER  $120 \text{ ha}$

3. A room is  $450 \text{ cm}$  by  $320 \text{ cm}$ . Find its area in  $\text{m}^2$ .

ANSWER  $14.4 \text{ m}^2$

4. A field is  $250 \text{ m}$  by  $400 \text{ m}$ . Find its area in hectares.

ANSWER  $10 \text{ ha}$

5. If lengths are halved, what happens to areas?

ANSWER They become a quarter.

6. Convert  $85\,000 \text{ cm}^2$  to  $\text{m}^2$ .

ANSWER  $8.5 \text{ m}^2$

7. A rectangle is  $1.5 \text{ km}$  by  $800 \text{ m}$ . Find its area in  $\text{km}^2$ .

ANSWER  $1.2 \text{ km}^2$

### Hard · stretch and reason

7 PROBLEMS

1. On a 1 : 10 000 map a park covers 12 cm<sup>2</sup>. Find its real area in hectares.

ANSWER  $12 \cdot 10\,000^2 = 1.2 \times 10^9 \text{ cm}^2 = 120\,000 \text{ m}^2 = 12 \text{ ha}$

2. A photograph is enlarged so that lengths are 4 times bigger. Its original area was 60 cm<sup>2</sup>. Find the new area.

ANSWER 960 cm<sup>2</sup>

3. A district of 45 km<sup>2</sup> is shown on a 1 : 50 000 map. Find its area on the map, in cm<sup>2</sup>.

ANSWER  $45 \times 10^{10} \text{ cm}^2 \div 2.5 \times 10^9 = 180 \text{ cm}^2$

4. A farm of 3.6 ha is a rectangle 240 m long. Find its width.

ANSWER 150 m

5. A tile is 25 cm square. How many are needed for a floor of 18 m<sup>2</sup>?

ANSWER 288

6. Two similar rectangles have sides in the ratio 2 : 5. The smaller has area 12 cm<sup>2</sup>. Find the larger.

ANSWER 75 cm<sup>2</sup>

7. Explain why the area conversion factor is the square of the length factor.

ANSWER Area is a product of two lengths; multiplying each of them by  $k$  multiplies the product by  $k^2$ .

## 07 Homework

### Homework — 5 problems

Geometry 8, Tema 60, pp. 144–146. Write the conversion factor before converting.

- Convert 6.5 m<sup>2</sup> to cm<sup>2</sup> and 120 000 cm<sup>2</sup> to m<sup>2</sup>.
- Convert 4.8 km<sup>2</sup> to hectares.
- A room is 520 cm by 380 cm. Find its area in m<sup>2</sup>.

4. *A field 300 m by 500 m — find its area in hectares and in  $\text{km}^2$ .*
5. *On a 1 : 20 000 map a lake covers 5  $\text{cm}^2$ . Find its real area in hectares.*